

JEE-Main-22-01-2026 (Memory Based)
[EVENING SHIFT]
Physics

Question: Find the dimensions of the expression $\frac{\epsilon_0 E}{T}$, where ϵ_0 , E and T are permittivity, electric field and time.

Options:

- (a) MLA^2
- (b) $MA^{-1}L$
- (c) AL^{-2}
- (d) AL

Answer: (c)

Solution:

$$\left[\frac{\epsilon_0 E}{T} \right] = \frac{C^2}{\cancel{Nm^2}} \times \frac{\cancel{C} \times T}{\cancel{m^2} T} = \frac{C}{m^2 T} = IL^{-2}$$

$$= AL^{-2}$$

Question: In an open organ pipe 3rd and 6th harmonic frequency differ by 3200 Hz. Find the length of organ pipe (speed of sound = 320 m/s)

Options:

- (a) 5 cm
- (b) 10 cm
- (c) 15 cm
- (d) 20 cm

Answer: (c)

Solution:

$$f_0 = \frac{nV}{2L_0}$$

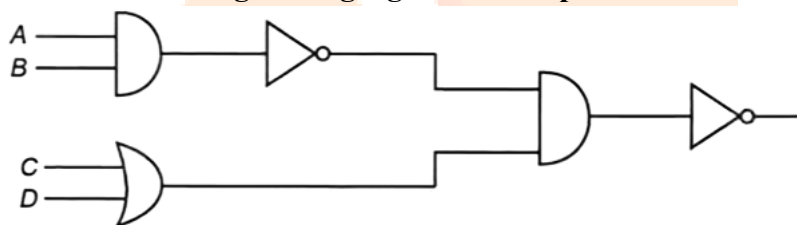
$$f_6 - f_3 = 3200 \text{ Hz}$$

$$\Rightarrow \frac{6V}{2L_0} - \frac{3V}{2L_0} = 3200$$

$$\Rightarrow \frac{3V}{2L_0} = 3200$$

$$\Rightarrow L_0 = \frac{1.5 \times 320}{\frac{3200}{10}} = 0.15 \text{ m} = 15 \text{ cm}$$

Question: For the given logic gate find output function.

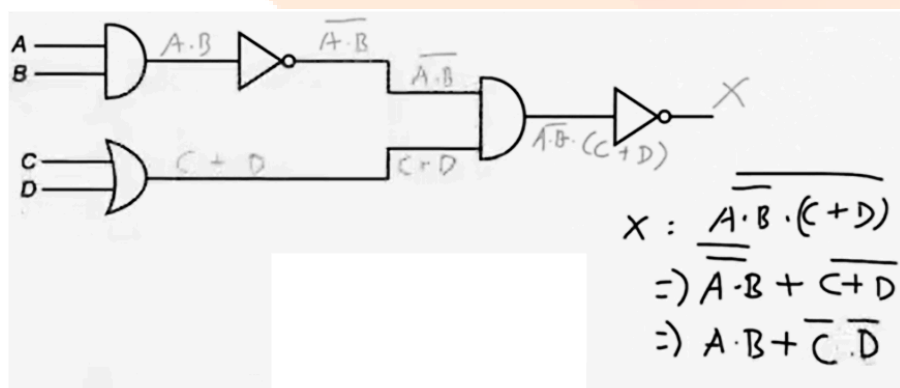


Options:

- (a) $\overline{A} \cdot \overline{B} + C + D$
- (b) $\overline{A} + \overline{B} + \overline{C} \cdot \overline{D}$
- (c) $AB + CD$
- (d) $AB + \overline{C} \cdot \overline{D}$

Answer: (d)

Solution:



Question: 3 small identical bubbles of water having same charge on each coalesce to form bigger bubble, Then the ratio of the potentials on one initial bubble & that on the resultant bigger bubble is:

Options:

(a) $1 : 3^{2/3}$

(b) $3^{2/3} : 1$

(c) $1 : 2^{2/3}$

(d) $1 : 3^{1/3}$

Answer: (a)

Solution:

$$V = \frac{Kq}{r}$$

$$V_{\text{new}} = \frac{K \cdot 3q}{r\sqrt{3}}$$

$$= \frac{Kq}{r} \cdot \sqrt{3}$$

$$= V \cdot \sqrt{3}$$

$$\frac{V}{V_{\text{new}}} = \frac{1}{\sqrt{3}}$$

$$q + q + q = 3q$$

$$n + n + n \Rightarrow N \dots \text{moles.}$$

$$\frac{P \cdot V}{RT} \times 3 = \frac{P_{\text{new}} V_{\text{new}}}{RT}$$

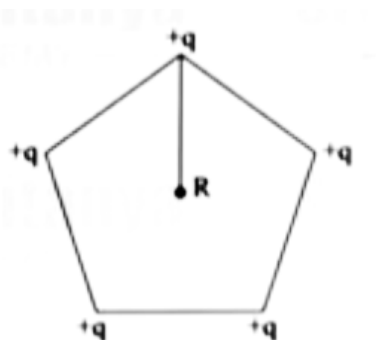
$$\frac{45}{r} \times \frac{4}{3} \pi r^3 \times 3 = \frac{45}{R} \times \frac{4}{3} \pi R^3$$

$$3r^2 = R^2$$

$$R = r\sqrt{3}$$

Question: Five positive charges each having charge q are placed at the vertices of a

pentagon as shown in the figure. The electric potential (V) & the electric field $\left(\vec{E}\right)$ at the center O of the pentagon due to the 5 positive charges are:-



Options:

(a) $V = 0, E = 0$

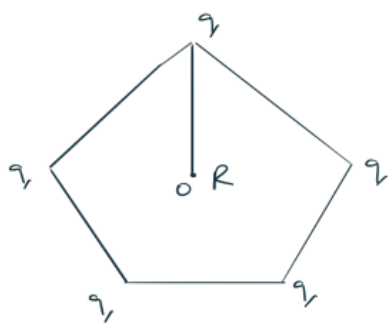
(b)

(c) $V = \frac{5a}{4\pi\epsilon_0 r}, \vec{E} = 0$

(d)

Answer: (c)

Solution:



Electric potential :-

due to each charge is $\frac{q}{4\pi\epsilon_0 r}$

due to 5 charges $V_o = \frac{5q}{4\pi\epsilon_0 r}$

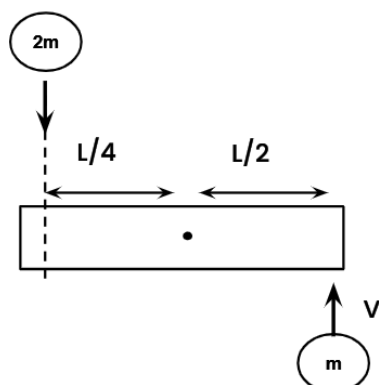
Electric field :-

As electric field is vector

quantity.

$$\vec{E}_o = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4 + \vec{E}_5 = 0 //$$

Question: Two balls of mass $2m$ and m collides with rod of mass m and length L as shown balls stick to the rod after collision. Find $\frac{V}{\omega}$ if rod is hinged at centre. ($L = 8\text{ m}$)

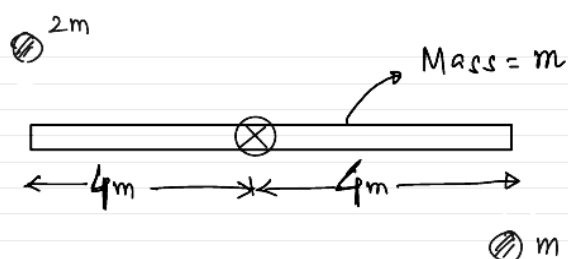


Options:

- (a) $11/2$
- (b) $11/3$
- (c) $40/9$
- (d) $9/4$

Answer: (c)

Solution:



Angular momentum of balls:-

$$L_i = (2m V \times 4) + (m V \times 4) = 12mV$$

Angular momentum of rod + balls after collision:-

$$L_f = \left[(2m \times 16) + (m \times 16) + \frac{1}{12} \times m \times 64 \right] \omega$$

$$= \left[48 + \frac{16}{3} \right] \omega m$$

$$= \frac{160 \omega m}{3}$$

Angular momentum Conservation: -

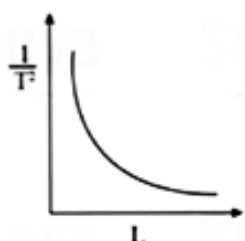
$$L_i = L_f$$

$$\Rightarrow 12mV = \frac{160 \omega m}{3}$$

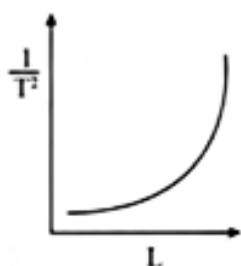
$$\Rightarrow \frac{V}{\omega} = \frac{160}{12 \times 3} = \frac{40}{9}$$

Question: Using a simple pendulum experiment g is determined by measuring its time period T . Which of the following plots represent correct relation b/w the pendulum length l & time period T .

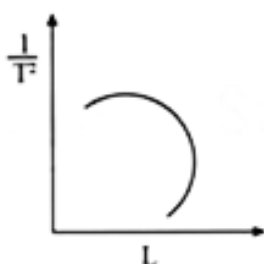
Options:



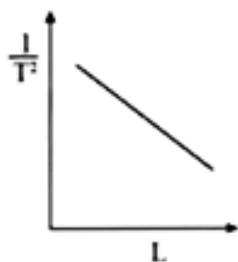
(a)



(b)



(c)



(d)

Answer: (a)

Solution:

$$T = 2\pi \sqrt{\frac{L}{g}}$$

SOBS

$$T^2 = 4\pi^2 \left(\frac{L}{g}\right)$$

$$\frac{1}{T^2} = \frac{g}{4\pi^2 L}$$

$$\frac{1}{T^2} \propto \frac{1}{L}$$

Rectangular hyperbola.

Question: A gas undergoes a process in which state variable changes from (1 atm, 60 ml, 27°C) to (P atm, 30 ml, 77°C) then P is

Options:

- (a) 3 atm
- (b) 5/4 atm
- (c) 7/3 atm
- (d) 4/3 atm

Answer: (c)

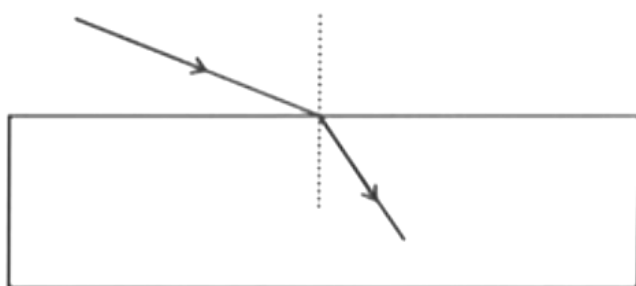
Solution:

<u>Initial state</u>	<u>Final state</u>
$P_1 = 1 \text{ atm}$	$P_2 = P \text{ atm}$
$V_1 = 60 \text{ ml}$	$V_2 = 30 \text{ ml}$
$T_1 = 27^\circ\text{C}$	$T_2 = 77^\circ\text{C}$
from ideal gas eq ⁿ	
$PV = nRT$	

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\frac{1 \times 60}{300} = \frac{P \times 30}{350} \Rightarrow P = \frac{7}{3} \text{ atm}$$

Question: A light ray incident on a slab of refractive index $\frac{3}{2}$. If 2 wavelength of refracted ray is 520 nm. Find wavelength of incident ray.

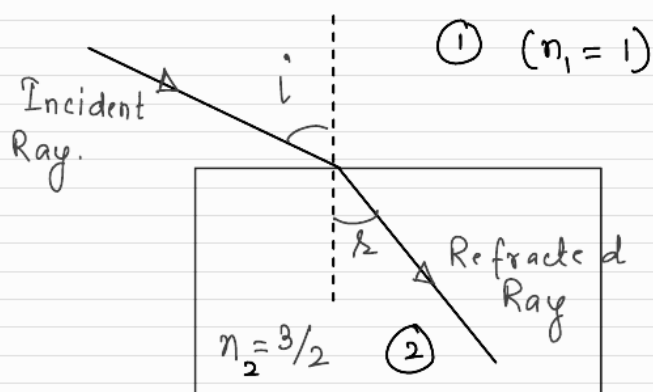


Options:

- (a) 460 nm
- (b) 780 nm
- (c) 360 nm
- (d) 560 nm

Answer: (b)

Solution:



$$\lambda_r = 520 \text{ nm} ; \lambda_i = ?$$

Snell's law:-

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} = \frac{v_1}{v_2} = \frac{\lambda_i}{\lambda_r}$$

$$\therefore \frac{3/2}{1} = \frac{\lambda_i}{520}$$

$$\Rightarrow \lambda_i = 1.5 \times 520 = 780 \text{ nm}$$

Correct option (2)

Question: Fig shows a circuit that contains 3 resistances (9Ω each) & two inductors (4 mH). The reading of ammeter at the moment switch K is turned ON, is _____ A.

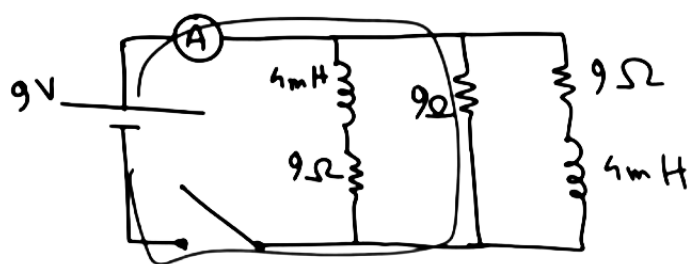


Options:

- (a) 3
- (b) 0
- (c) 2
- (d) 1

Answer: (d)

Solution:



$$9/9 = 1$$

Question: Which of the following is/are true for YDSE experiment?

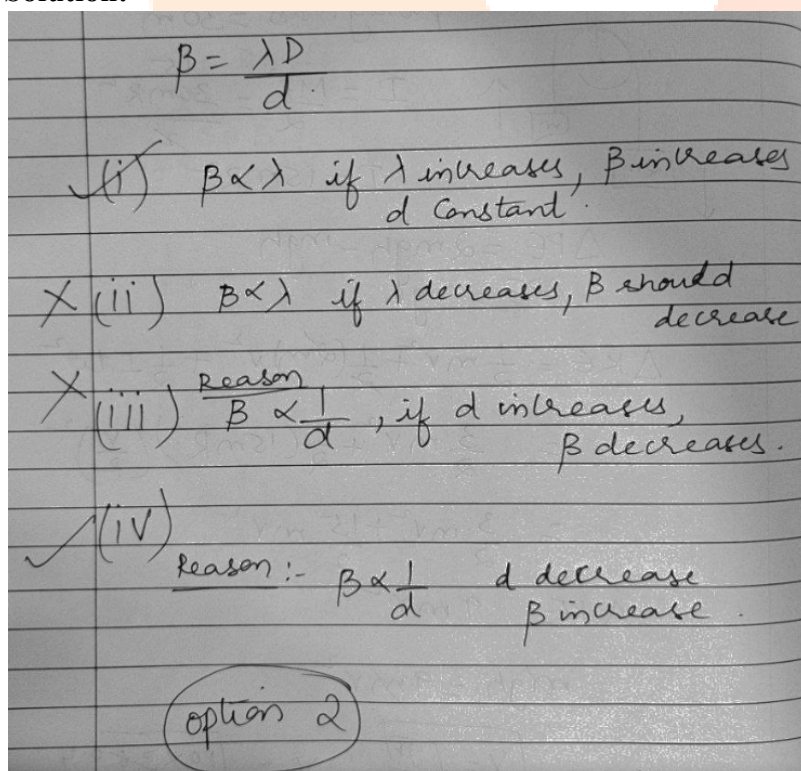
- i. Fringe width increases if distance between slits width is constant & wavelength increased.
- ii. Fringe width increases if distance between slits is constant & wavelength decreased.
- iii. Fringe width increases if distance between slits is increased & wavelength constant.
- iv. Fringe width increases if distance between slits is decreased & wavelength constant.

Options:

- (a) i, iii
- (b) i, iv
- (c) ii, iii
- (d) ii, iv

Answer: (b)

Solution:



Question: A pulley has mass 3 cm there are two blocks of masses m & $2M$. Find speed of $2m$ when it descends by distance 3.6 meter.

Options:

- (a) 4
- (b) 2
- (c) 8
- (d) 12

Answer: (b)

Solution:

pulley mass = $30m$

$$I = \frac{MR^2}{2} = \frac{15mR^2}{2}$$

$$I = 15mR^2$$

$$\Delta PE = 2mgh - mgh$$

$$\Delta PE = mgh$$

$$\Delta KE = \frac{1}{2}mv^2 + \frac{1}{2}(2m)v^2 + \frac{1}{2}I\omega^2$$

$$= \frac{3}{2}mv^2 + \frac{1}{2}(15mR^2)\left(\frac{v}{R}\right)^2$$

$$= \frac{3}{2}mv^2 + \frac{15}{2}mv^2$$

$$\Delta KE = 9mv^2$$

$$mgh = 9mv^2$$

$$v = \sqrt{\frac{gh}{9}}$$

$$v = \sqrt{\frac{10 \times 3.6 \times 0.4}{9}}$$

$$v = 2 \text{ m/s}$$

Question: The smallest wavelength of Lyman series is λ_0 . The difference between the largest wavelength of Paschen and Balmer series is nearly.

Options:

- (a) $\frac{464\lambda_0}{5}$
- (b) $\frac{566\lambda_0}{54}$
- (c) $\frac{468\lambda_0}{35}$
- (d) $534\lambda_0$

Answer: (c)

Solution:

Smallest wavelength of Lyman Series = λ_0

$$n_L = 1; \quad n_H = \infty$$

$$\frac{1}{\lambda_0} = R \left[\frac{1}{1^2} - \frac{1}{\infty^2} \right] = R$$

$$\lambda_0 = \frac{1}{R} \quad \text{--- (1)}$$

for largest wavelength of Paschen Series:-

$$n_L = 3; \quad n_H = 4$$

$$\frac{1}{\lambda_P} = R \left[\frac{1}{3^2} - \frac{1}{4^2} \right] = \frac{7R}{144}$$

for largest wavelength of Balmer Series:-

$$\frac{1}{\lambda_B} = R \left[\frac{1}{2^2} - \frac{1}{3^2} \right] = \frac{5R}{36}$$

$$\therefore \lambda_P - \lambda_B = \frac{144}{7R} - \frac{36}{5R} = \frac{468\lambda_0}{35}$$

Question: In case of capillary action if surface tension of liquid, radius of capillary and density of liquid decreases to 1% then percentage change in height of liquid level inside capillary:

Options:

- (a) 1% increase
- (b) 1% decrease
- (c) 2% increase
- (d) 2% decrease

Answer: (a)

Solution:

Capillary rise "h" in a capillary Tube is:-

$$h_1 = \frac{2 T_1 \cos \theta}{\rho_1 g r_1}$$

$$T_2 = T_1 - 1\% T_1 = \frac{99}{100} T_1 = 0.99 T_1$$

$$\rho_2 = 0.99 \rho_1$$

$$r_2 = 0.99 r_1$$

$$\therefore h_2 = \frac{2 T_2 \cos \theta}{\rho_2 g r_2} = \frac{2 \times 0.99 T_1 \cos \theta}{0.99 \rho_1 g (0.99 r_1)}$$

$$= \frac{1}{0.99} \times h_1$$

$$\frac{h_2}{h_1} = \frac{1}{0.99} = \frac{100}{99}$$

$$\therefore \text{Increase} = \left(\frac{h_2}{h_1} - 1 \right) \times 100 = \frac{100}{99} = 1.01\%$$

Correct option (1).

Question: Statement-1: Work done by $\vec{F}$ from $\vec{r}_1$ to $\vec{r}_2$ is given as

$$W = - \int_{r_1}^{r_2} \vec{F} \cdot d\vec{r}, \quad \text{if } \vec{F} \text{ is conservative.}$$

Statement -1: There are ∞ ways through which we can go from $\vec{r}_1$ to $\vec{r}_2$ and work done for each case will be different for conservative force.

Options:

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is correct explanation of Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT correct explanation of Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 and Statement-2 both are False.

Answer: (d)

Solution:

statement 1 :- false

work done is always

$$\int \vec{F} \cdot d\vec{r} \text{ always}$$

statement 2 :- false

work done by a conservative

force does not depends on path.

Question: A laser beam has intensity of $4.0 \times 10^{14} \text{ W/m}^2$. The amplitude of magnetic field associated with beam is _____ T.

Options:

- (a) 1.83
- (b) 5.5
- (c) 18.3
- (d) 2.0

Answer: (a)

Solution:

$$I = \frac{1}{2} \frac{B_0^2}{\mu_0} c$$

$$B_0 = \sqrt{\frac{2\mu_0 I}{c}} = \sqrt{\frac{2 \times 4\pi \times 10^{-7} \times 4 \times 10^{14}}{3 \times 10^8}}$$

Question: Statement-1: Kinetic energy of system

$$= \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 + \dots + \frac{1}{2} m_n v_n^2.$$

Statement-2: Kinetic energy of system = Kinetic energy of centre of mass + Kinetic energy with respect to centre of mass.

Options:

- (a) Statement I is true. Statement II is true
- (b) Statement I is true. Statement II is false
- (c) Statement I is false. Statement II is true

(d) Statement I is false. Statement II is false

Answer: (a)

Solution:

Q:-

$m_1 \rightarrow v_1$ $M = m_1 + m_2$ $m_2 \rightarrow v_2$
 $\bullet \rightarrow v_{cm}$

$$KE_{sys} = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \quad \text{--- (1)}$$

$$\begin{aligned}
 KE_{cm} &= \frac{1}{2} M v_{cm}^2 = \frac{1}{2} (m_1 + m_2) \left[\frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} \right]^2 \\
 &= \frac{1}{2} \frac{m_1^2 v_1^2 + m_2^2 v_2^2 + 2 m_1 m_2 v_1 v_2}{(m_1 + m_2)} \quad \text{--- (2)}
 \end{aligned}$$

With respect to center of mass:-

$$\begin{aligned}
 KE_1 &= \frac{1}{2} m_1 (v_1 - v_{cm})^2 = \frac{1}{2} m_1 \left[v_1 - \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} \right]^2 \\
 KE_2 &= \frac{1}{2} m_2 (v_2 - v_{cm})^2 = \frac{1}{2} m_2 \left[v_2 - \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} \right]^2 \\
 KE_1 + KE_2 &= \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} (v_1 - v_2)^2 \quad \text{--- (3)}
 \end{aligned}$$

Adding (2) & (3) we get:-

$$KE_{sys} = KE_{cm} + \text{K.E. of System with respect to com.}$$

Correct option (1).

Question: A capacitor of capacitance $10 \mu\text{F}$ is connected with a battery 6 V . Now battery is disconnected and another uncharged capacitor of capacitance $20 \mu\text{F}$ is connected to the capacitor. Find charge on $20 \mu\text{F}$ capacitor.

Options:

- (a) $\frac{30}{4} \mu\text{C}$
 (b) $10 \mu\text{C}$

- (c) $\frac{20}{3} \mu C$
 (d) $40 \mu C$

Answer: (d)

Solution:

$Q_1 = C_1 V$
 $= 10 \times 6$
 $Q_1 = 60 \mu C$

$C_T = C_1 + C_2$
 $= 10 + 20$
 $C_T = 30 \mu F$

$V_{common} = \frac{Q_T}{C_T} = \frac{60 \mu C}{30 \mu F} = 2 V$

$Q_2 = C_2 \times V_c$
 $= 20 \mu F \times 2$
 $Q_2 = 40 \mu C$

option 4

Question: For the given statements below mark the correct option.

Statement-I: Work done by conservative force f , from $\vec{r}_1$ to $\vec{r}_2$ is given by

$$W = - \int_{r_1}^{r_2} \vec{F} \cdot d\vec{r}$$

Statement-II: Work done by conservative force is path dependent.

Options:

- (a) Statement-I and statement-II is true
 (b) Statement-I is false and statement-II is true
 (c) Statement-I is true and statement-II is false
 (d) Statement-I and statement-II is false

Answer: (d)

Solution:

Question: Electromagnetic wave with intensity $I = 4 \times 10^{14}$ watt/m² is propagating in free space. Find the amplitude of magnetic field B_0 .
(Use $c = 3 \times 10^8$ m/s, $\epsilon_0 = 8.85 \times 10^{-12}$ C²/N.m²).

Options:

- (a) 1.83 Tesla
- (b) 0.5 Tesla
- (c) 4.5 Tesla
- (d) 1 Tesla

Answer: (a)

Solution:

$$I = \frac{1}{2} \epsilon_0 E_0^2 c \quad \left| \quad I = \frac{1}{2} \epsilon_0 c^3 B_0^2 \right| \quad \left| \quad B_0^2 = \frac{2 \times 4 \times 10^{14}}{8.85 \times 10^{-12} \times (3 \times 10^8)^3} \right|$$

$$c = \frac{E_0}{B_0} \Rightarrow E_0 = c B_0 \quad \left| \quad B_0^2 = \frac{2I}{\epsilon_0 c} \right| \quad \left| \quad B_0 = 1.83 \text{ T} \right|$$

Question: A cylindrical conductor of Length 2 m and Area of cross section, 0.2 mm² carries an electric current of 1.6 A when its ends are connected to 2V battery. Mobility of electrons in the conductor is $\alpha \times 10^{-3}$ m²/v - s. The value of α is.
[Electron concentration = 5×10^{28} /m³ and Electron charge = 1.6×10^{-19} C].

Options:

- (a) 1
- (b) 4
- (c) 5
- (d) 10

Answer: (a)

Solution:

$$E = \frac{V}{L}$$

$$i = neAv_d$$

$$\Rightarrow v_d = \frac{i}{neA}$$

$$\mu = \frac{v_d}{E}$$

$$\mu = \frac{iL}{neAV}$$

$$= 1 \times 10^{-3}$$

$$1.0 \times 10^{-3}$$

Question: The wavelength of light when it is passing through water is 540 nm. The refractive index of water is $\frac{4}{3}$. The wavelength of the same light when it is passing through a transparent medium having refractive index of $\frac{3}{2}$ is.

Options:

- (a) 840 nm
- (b) 480 nm
- (c) 400 nm
- (d) 356 nm

Answer: (b)

Solution:

$$\lambda = \frac{\lambda_0}{n} \Rightarrow \lambda_0 = n \lambda$$

$$\frac{4}{3} \times 540 = \frac{3}{2} \times \lambda'$$

$$\Rightarrow \lambda' = \frac{4 \times 2}{3} \times 540 = 480$$

Question: S-I:- A satellite is moving around earth in the orbit very close to the earth's surface. The time period of the revolution depends on the density of the earth.

$$T = 2\pi \sqrt{\frac{R_e}{g}}, \text{ where } R_e \text{ is}$$

S-II:- The time period of the revolution of the satellite is the radius of the earth and g is the acceleration due to gravity.

Options:

- (a) S-I is true but S-II is false.
- (b) Both S-I and S-II are true.
- (c) S-I is false but S-II is true.
- (d) Both S-I and S-II are false.

Answer: (b)

Solution:

$$T = \frac{2\pi R^{3/2}}{\sqrt{GM}}$$

$$g = \frac{GM}{R_e^2}$$



JEE-Main-22-01-2026 (Memory Based)
[EVENING SHIFT]
Chemistry

Question: The correct order of electron gain enthalpy (magnitude only) for group 16 elements is

Options:

- (a) $\text{Te} > \text{Se} > \text{S} > \text{O}$
- (b) $\text{S} > \text{Se} > \text{Te} > \text{O}$
- (c) $\text{O} > \text{S} > \text{Se} > \text{Te}$
- (d) $\text{S} > \text{O} > \text{Se} > \text{Te}$

Answer: (b)

Question: 100 g 98% by weight H_2SO_4 is mixed with 100g 49% by weight H_2SO_4 . Mole fraction of H_2SO_4 in solution is

Options:

- (a) 0.9
- (b) 0.1
- (c) 0.67
- (d) 0.33

Answer: (d)

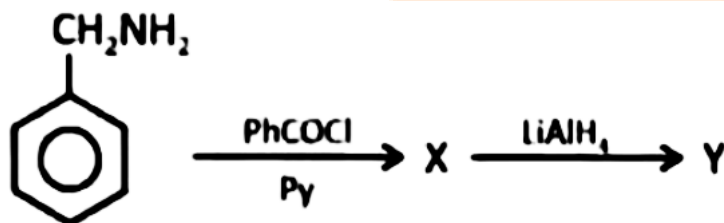
Question: Correct order of ionisation enthalpy is

Options:

- (a) $\text{F} > \text{Cl} > \text{Cl}^- > \text{F}^-$
- (b) $\text{F}^- > \text{Cl}^- > \text{F} > \text{Cl}$
- (c) $\text{Cl} > \text{F} > \text{Cl}^- > \text{F}^-$
- (d) $\text{F} > \text{Cl} > \text{F}^- > \text{Cl}^-$

Answer: (a)

Question: Consider the following reaction.



The correct structure of Y is

Options:

- (a) $\text{PhCH}_2\text{NHCOPh}$
- (b) $\text{Ph-CH}_2\text{NHCH}_2\text{Ph}$
- (c) $\text{PhNH}_2\text{CH}_2\text{Ph}$
- (d) PhCH_3

Answer: (b)

Question: Which of the following is a mixed oxide?

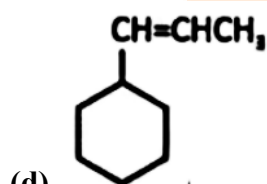
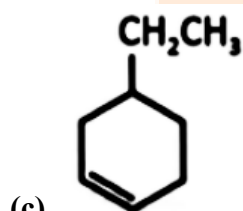
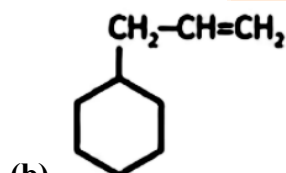
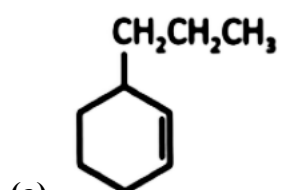
Options:

- (a) Fe_2O_3
- (b) PbO_2
- (c) Pb_3O_4
- (d) BaO_2

Answer: (c)

Question: An alkene on reductive ozonolysis gives methanal as one of the products. It's structure is

Options:



Answer: (b)

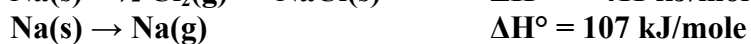
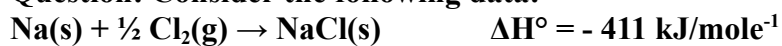
Question: Which of following is basic buffer?

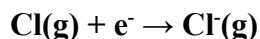
Options:

- (a) $\text{NaOH} + \text{CH}_3\text{COONa}$
- (b) $\text{NaOH} + \text{Na}_2\text{SO}_4$
- (c) $\text{K}_2\text{SO}_4 + \text{H}_2\text{SO}_4$
- (d) $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$

Answer: (d)

Question: Consider the following data:





$$\Delta H^\circ = -355 \text{ kJ/mole}$$



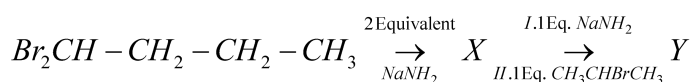
$$\Delta H^\circ = 502 \text{ kJ/mole}^{-1}$$

Find out lattice energy of NaCl(s)

Options:

- (a) -786 kJ mole⁻¹
- (b) -628 kJ mol⁻¹
- (c) -428 kJ mole⁻¹
- (d) -393 kJ mole⁻¹

Answer: (a)



Question:

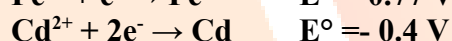
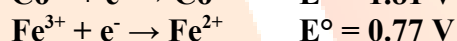
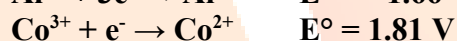
Options:

- (a) 2-methyl hex-3-yne
- (b) 5-methyl hex-2-yne
- (c) Isopropyl but-1-yne
- (d) 2-Methyl hex-2-yne

Answer: (a)

Question: Which of the following will behave as best reducing agent?

Given



Options:

- (a) Al
- (b) Co²⁺
- (c) Fe²⁺
- (d) Cd

Answer: (a)

Question: Given below are two statements.

Statement I: First ionisation enthalpy of Cr is greater than Mn.

Statement II: Second and third ionisation enthalpy of Cr is less than that of Mn.

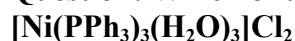
In the light of above statements, choose the correct option.

Options:

- (a) Both statement I and statement II are correct
- (b) Both statement I and statement II are incorrect
- (c) Statement I is correct but statement II is incorrect
- (d) Statement I is incorrect but statement II is correct

Answer: (b)

Question: Which of the following is the correct IUPAC name of complex?



Options:

- (a) Triaquatris(triphenylphosphine)nickel(II) chloride
- (b) Tris(triphenylphosphine)triaquanickel(II) chloride
- (c) Triaquatris(triphenylphosphine)nickelate(II) chloride
- (d) Triaquatris(triphenylphosphine)nickel(III) chloride

Answer: (a)

Question: Match the two columns: Glucose + 'X' → 'P'

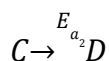
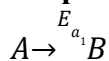
List-I (Reagent-X)	List-II (Product-P)
A) Br ₂ /water	i) Glucosamine
B) Acetic anhydride (excess)	ii) Saccharic acid
C) Conc. HNO ₃	iii) Glucose pentaacetate
D) NH ₂ OH	iv) Gluconic acid

Options:

- (a) A-iv, B-ii, C-iii, D-i
 (b) A-ii, B-iv, C-iii, D-i
 (c) A-ii, B-iii, C-iv, D-i
 (d) A-iv, B-iii, C-ii, D-i

Answer: (d)

Question: Consider two reactions having same pre-exponential factor (A) at same temperature (T).



$$E_{a_1} = 5E_{a_2}$$

Find out correct expression?

Options:

(a) $\frac{k_1}{k_2} = e^{-\frac{E_{a_2}}{RT}}$

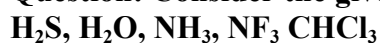
(b) $\frac{k_1}{k_2} = e^{-\frac{4E_{a_1}}{RT}}$

(c) $\frac{k_1}{k_2} = e^{-\frac{4E_{a_1}}{5RT}}$

(d) $\frac{k_1}{k_2} = e^{-\frac{4E_{a_2}}{5RT}}$

Answer: (c)

Question: Consider the given species



The number of lone pair on central atom which lowest dipole moment.

Options:

- (a) 0
 (b) 1

(c) 2

(d) 3

Answer: (b)

Question: Consider the statements below

Statement-I: BCl_3 is covalent in natureStatement-II: BCl_3 undergo hydrolysis to form $[\text{B}(\text{OH})_4]^-$ and BH_3^+

In the light of above statements choose the correct option.

Options:

(a) Statement-I and statement-II both are correct

(b) Statement-I and Statement-II both are incorrect

(c) Statement-I correct statement-II incorrect

(d) Statement-I incorrect statement-II correct

Answer: (c)

Question: 36 g of A reacts with 54 g of B to form AB_2 , if molar mass of A and B is respectively 60 and 80, then choose correct option from following.

Options:

(a) Limiting Reagent is A

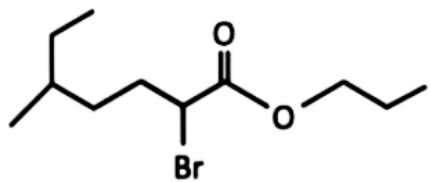
(b) 90 g of AB_2 formed

(c) Limiting Reagent is B

(d) 50 g of AB_2 formed

Answer: (c)

Question: The correct IUPAC nomenclature of the following compound is



Options:

(a) Propyl-2-bromo-6-methyl heptanoate

(b) 2-Bromo-5-methyl-1-propyl heptanoate

(c) Propyl-2-bromo-5-ethyl hexanoate

(d) Propyl-2-bromo-5-methyl heptanoate

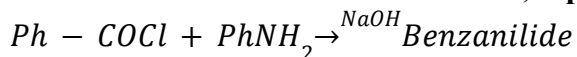
Answer: (d)

Question: Volume ratio of decimolar NH_4OH and decimolar HCl to give a solution of $\text{pH} = 9.25$ at 25°C is $x : 1$. Find x . pK_b of $\text{NH}_4\text{OH} = 4.75$

Options:

Answer: (2)

Question: 5.8 g Aniline is converted into benzanilide with some reaction sequences. Calculate mass of benzanilide formed, if percentage yield of reaction is 82%



Options:

Answer: (10)

Question: A cycloalkene (X) is treated with Br_2 and compound (Y) is formed with C : Br ratio 3 : 1. One mole of X required 1 mol of Br_2 . Find composition (percentage) of 'Br' in Y compound (percentage)

Options:

Answer: (66)



JEE-Main-22-01-2026 (Memory Based)

[EVENING SHIFT]

Maths

Question: If complex number $Z_1, Z_2, \dots, Z_n$ satisfy the equation

$$4Z^2 + \bar{Z} = 0, \text{ then } \sum_{i=1}^n |Z_i|^2 \text{ is equal to}$$

Options:

- (a) $\frac{3}{16}$
- (b) $\frac{3}{64}$
- (c) $\frac{64}{9}$
- (d) $\frac{1}{16}$

Answer: (a)

$$4z^2 + \bar{z} = 0 \dots \dots (1)$$

$$\text{If } z = 0$$

$$\Rightarrow |z| = 0$$

$$\text{If } z \neq 0$$

Multiply z on (1)

$$4z^3 + z \cdot \bar{z} = 0$$

$$4|z|^3 = |z|^2$$

$$\Rightarrow |z| = \frac{1}{4}$$

Put

$$4z^3 = -\frac{1}{16}$$

$$z^3 = -\frac{1}{64}$$

$$\text{possible } z = \frac{-1}{4}, \frac{-1}{4}\omega, \frac{-1}{4}\omega^2$$

$$\sum |z_i|^2 = 0 + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} = \frac{3}{16}$$

Question: Let α, β be the roots of quadratic equation $12x^2 - 20x + 3\lambda = 0$, $\lambda \in \mathbb{Z}$. If

$\frac{1}{2} \leq |\beta - \alpha| \leq \frac{3}{2}$ then the sum of all possible values of λ is

Answer: (3)

$$12x^2 - 20x + 3\lambda = 0$$

$$a = 12, b = -20, c = 3\lambda$$

$$\alpha + \beta = \frac{-20}{12} = \frac{5}{3}$$

$$\alpha\beta = \frac{3\lambda}{12} = \frac{\lambda}{4}$$

$$\frac{1}{2} \leq |\beta - \alpha| \leq \frac{3}{2} \dots\dots(2)$$

$$\begin{aligned} (\beta - \alpha)^2 &= (\beta + \alpha)^2 - 4\alpha\beta \\ &= \left(\frac{5}{3}\right)^2 - 4\left(\frac{\lambda}{4}\right) \end{aligned}$$

$$(\beta - \alpha)^2 = \frac{25}{9} - \lambda$$

$$|\beta - \alpha| = \sqrt{\frac{25}{9} - \lambda} \dots\dots(2)$$

$$(1) \& (2) \quad \frac{1}{2} \leq \sqrt{\frac{25}{9} - \lambda} \leq \frac{3}{2}$$

$$S.B.S. \quad \frac{1}{4} \leq \frac{25}{9} - \lambda \leq \frac{9}{4}$$

$$-\frac{91}{36} \leq -\lambda \leq -\frac{19}{36}$$

$$\frac{19}{36} \leq \lambda \leq \frac{91}{36}$$

$$\approx 0.527 \leq \lambda \leq 2.527$$

$$= 1, 2; \quad \sum \lambda = 1 + 2 = 3$$

Question: The number of elements in the relation $R = \{(x, y) : 4x^2 + y^2 < 52, x, y, \in \mathbb{Z}\}$ is

Answer: (77)

$$x = 0, y^2 < 52, y = 0, \pm 1, \pm 2, \pm 3, \pm 4, \pm 5, \pm 6, \pm 7$$

$$x = \pm 1 \quad y^2 < 48$$

$$x = \pm 2 \quad y^2 < 36$$

$$x = \pm 3 \quad y^2 < 16$$

$$2 \times 13 = 26$$

$$2 \times 11 = 22$$

$$2 \times 7 = 14$$

$$15 + 26 + 22 + 14 = 77$$

Question: Area enclosed by $4x^2 + y^2 \leq 8$ and $y^2 \leq 4x$ (in square unit) is

Options:

(a) $\left(\pi + \frac{4}{3}\right)$ sq. units

(b) $\left(\pi - \frac{4}{3}\right)$ sq. units

(c) $\left(\pi + \frac{2}{3}\right)$ sq. units

(d) $\left(\pi - \frac{2}{3}\right)$ sq. units

Answer: (c)

$$4x^2 + y^2 \leq 8$$

$$\frac{x^2}{2} + \frac{y^2}{8} \leq 1 \dots (1)$$

$$y^2 \leq 4x$$

Find intersecting

$$4x^2 + 4x = 8$$

$$x^2 + x - 2 = 0$$

$$(x + 2)(x - 1) = 0$$

$$x = 1, y = \pm 2$$

$$A = 2 \left[\int_0^1 2\sqrt{x} \cdot dx + \int_1^{\sqrt{2}} 2\sqrt{2-x^2} \cdot dx \right]$$

$$= 2 \left[2 \left[\frac{2}{3} x^{\frac{3}{2}} \right]_0^1 + 2 \left(\frac{x}{2} \right) \sqrt{2-x^2} + \frac{2}{2} \sin^{-1} \left(\frac{x}{\sqrt{2}} \right) \right]_1^{\sqrt{2}}$$

$$2 \left[2 \times \frac{2}{3} (1) + 2 \left[\frac{\pi}{2} - \left(\frac{1}{2} \right) + \frac{\pi}{4} \right] \right]$$

$$2 \left[\frac{4}{3} + 2 \left[\frac{\pi}{4} - \frac{1}{2} \right] \right]$$

$$A = \left[\frac{8}{3} + 2 \left(\frac{\pi}{2} - 1 \right) \right]$$

$$A = \pi + \frac{2}{3}$$

Question: If the mean deviation about the median of the numbers $k, 2k, 3k, \dots, 1000k$ is 500, then k^2 is equal to

Answer: (10^3)

For Mean Deviation about median for even no. of terms ($= n$) in A. P

$$\text{is } \left(\frac{k \cdot n}{4} \right) \quad k = \text{common difference}$$

$$= \left(\frac{1000}{4} \right) k \quad k = d$$

no. of terms

$$500 = \frac{1000}{4} k \quad n = 10^3$$

$$K = 2$$

$$\boxed{K^2 = 4}$$

Question: $x - ny + z = 6$

$$x - (n - 2)y + (n + 1)z = 8$$

$$(n - 1)y + z = 1$$

Let $n =$ number on the dies when rolled randomly then p (that system equation has unique solution) $= k/6$, then sum of value of k and all possible value of n is

Options:

(a) 22

(b) 21

(c) 20

(d) 24

Answer: (d)

$\therefore$ Unique solution, $\Delta \neq 0$

$$\Delta = \begin{vmatrix} 1 & -n & 1 \\ 1 & -(n-2) & (n+1) \\ 0 & (n-1) & 1 \end{vmatrix} \neq 0$$

$$\text{lets take } \Delta = 0; \{-(n-2) - (n-1)(n+1)\} - 1\{-n - (n-1)\}$$

$$-n + 2 - n^2 + 1 + n + n - 1 \quad n^2 - 1 \quad -n - n + 1$$

$$\Delta = -n^2 + n + 2 = 0 \Rightarrow n^2 - n - 2 = 0$$

$$(n-2)(n+1) = 0$$

$$n = -1 \text{ \& } 2$$

$\therefore$ system has unique solution ($\Delta \neq 0$) except for $n = -1$ & 2

$$\therefore n = \{1, 3, 4, 5, 6\}, k = 5 \text{ value}$$

$$\text{Sum} = k + 1 + 3 + 4 + 5 + 6 = 5 + 1 + 3 + 4 + 5 + 6$$

$$= 24$$

Question: If $P(10, 2\sqrt{15})$ lie on hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ & length of latus rectum is 8 then square of Area $\triangle P S_1 S_2$

Answer: (300)

$$\frac{2b^2}{a} = 8$$

$$b^2 = 4a$$

$$\frac{100}{a^2} - \frac{60}{b^2} = 1 \Rightarrow \frac{100}{a^2} - \frac{60}{4a} = 1 \Rightarrow \frac{100}{a^2} - \frac{15}{a} = 1$$

$$\Rightarrow 100 - 15a = a^2 \Rightarrow a^2 + 15a = 100 \Rightarrow a = 5$$

$$b^2 = 20$$

$$\text{Area} = \frac{1}{2} \cdot 2ae \cdot 2\sqrt{15} - \sqrt{a^2 - b^2} \cdot 2\sqrt{15}$$

$$= \sqrt{5} \times 2\sqrt{15}, \text{ square of area} = 300$$

$$\cos(\alpha + \beta) = -\frac{1}{10} \text{ and } \sin(\alpha - \beta) = \frac{3}{8} \text{ where}$$

$$0 < \alpha < \frac{\pi}{3} \text{ \& } 0 < \beta < \frac{\pi}{4} \text{ if } \tan 2\alpha = \frac{3(1-\gamma\sqrt{5})}{\sqrt{11}(s+\sqrt{5})}, \gamma, s \in N,$$

Question:

then $r + s$ is equal to _____

Answer: (20)

$$\cos(\alpha + \beta) = -\frac{1}{10} \in II$$

$$\sin(\alpha - \beta) = \frac{3}{8} \in I$$

$$\sin(\alpha + \beta) = \frac{3\sqrt{11}}{10} \in II$$

$$\cos(\alpha - \beta) = \frac{\sqrt{55}}{8} \in II$$

$$\sin 2\alpha = \sin((\alpha + \beta) + (\alpha - \beta))$$

$$\sin(\alpha + \beta) \cos(\alpha - \beta) + \cos(\alpha + \beta) \sin(\alpha - \beta)$$

$$= \frac{33\sqrt{5}-3}{80}$$

$$\cos 2\alpha = \cos((\alpha + \beta) + (\alpha - \beta))$$

$$= -\sqrt{11} \frac{(\sqrt{5}+9)}{80}$$

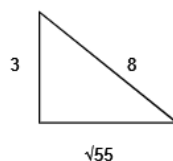
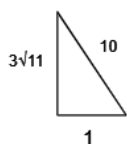
$$\tan 2\alpha = \frac{\sin 2\alpha}{\cos 2\alpha}$$

$$= \frac{3(-1+11\sqrt{5})}{-\sqrt{11}(9+\sqrt{5})}$$

$$= \frac{3(1-11\sqrt{5})}{\sqrt{11}(9+\sqrt{5})}$$

$$r = 11, s = 9$$

$$r + s = 20$$



Question: If $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ **is a Solution of Systems of equation** $AX = B$ **where**

$Adj A = \begin{pmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{pmatrix}$ **\& B =** $\begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$ **then** $|x + y + z|$ **is equal to**

Options:

- (a) 3
- (b) 2
- (c) $\frac{3}{2}$
- (d) 1

Answer: (b)

$$|Adj A| = 4(0 + 10) - 2(-20) + 2 \times 10$$

$$= 40 + 40 + 20 = 100 \Rightarrow |A| = 10$$

$$X = \frac{1}{10} \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$$

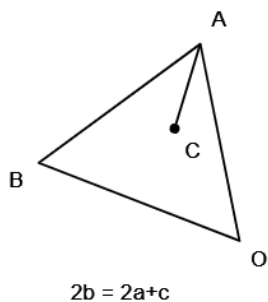
$$= \frac{1}{10} \begin{bmatrix} 16 + 0 + 4 \\ -20 + 0 + 10 \\ 4 + 0 + 6 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 20 \\ -10 \\ 10 \end{bmatrix}$$

$$|x + y + z| = 2$$

Question: Among the statements

S₁: If A(5,-1) and B(-2,3) are two vertices of a triangle, whose orthocenter is (0,0), then its third vertex is (-4,-7)

S₂: If positive numbers 2a, b, c are three consecutive, in terms of an A.P, then the lines $ax + by + c = 0$ are concurrent at (2,-2)



Options:

- (a) both are correct
- (b) only S₁ is correct
- (c) both are incorrect
- (d) only S₂ is correct

Answer: (d)

$$m_{BC} = \frac{10}{2} = 5$$

$$m_{AO} = \frac{-1}{5}$$

$$m_{AC} = \frac{6}{9} = \frac{2}{3}$$

$$m_{BO} = \frac{-3}{2}$$

S_1 is correct

$$2a + c = 2b \Rightarrow 2a - 2b + c = 0$$

$$ax + by + c = 0$$

$$(x, y) = (2, -2)$$

S_2 is correct

